MAXIMILIAN LIPSKI

Graphics Programmer | Vulkan & D3D12 | C++

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Summary

Computer Science master's student specializing in graphics programming and real-time rendering. Experienced in C++, Vulkan, and Direct3D 12, with projects spanning a custom rendering hardware interface, a physically based renderer, and a 3D game engine. Focused on rendering architecture, GPU programming, and performance optimization.

Experience

TU Dresden - ScaDS.AI — http://scads.ai — Dresden, DE

Student Researcher — 11/2025 - 04/2026

- Researched machine learning approaches to accelerate geometric rendering of geometric objects
- Developed and trained neural networks for geometric approximation problems using CUDA-enabled machine learning workflows
- Investigated neural representations of Hermite spline tubes as signed distance functions
- Trained a neural network for ray-intersection with Hermite spline tubes
- Designed and evaluated experiments to analyze the accuracy and performance of learned geometric approximations
- Achieved promising results for fixed primitives while identifying parameterization as a major challenge or generalization

Education

Technical University Dresden — Dresden, DE

Master of Science, Computer Science — 10/2024 - 01/2027

- Grade: 1.6
- Thesis: Rendering of transparent lines using hardware-accelerated linear swept sphere primitives
- Active, Expected Graduation: 01/2027

Heinrich-Heine University — Düsseldorf, DE

Bachelor of Science, Computer Science — 10/2020 - 09/2024

- Grade: 1.9
- Thesis: Development of a Skew-Heap Data Structure for Persistent Memory

Skills

Programming: C/C++ · GLSL · HLSL · CUDA · SPIR-V · Multithreading

Graphics APIs: Vulkan · Direct3D 12 · OpenGL

Tools: RenderDoc · NVIDIA Nsight Graphics · NVIDIA Nsight Systems · Git · CMake · Premake · Unreal Engine 5

Platforms: Linux · Windows

Research: Machine learning · Neural networks

Projects

Iryven -3D Game Engine

- Developing a C++20 engine with Vulkan and D3D12 rendering through my custom Velos RHI
- Implemented meshlet rendering, Hi-Z occlusion culling, and frame-graph scheduling
- Integrated Flecs ECS, Box3D physics, asynchronous asset loading and an ImGui scene editor

Rodan - Real-Time Rendering Engine

- Built a renderer with physically based shading, image-based lighting, shadow mapping and HDR tone mapping
- Added glTF loading with transmission and volume material support
- Profiled rendering with NVIDIA Nsight Graphics and RenderDoc; achieved approximately 400 FPS in Sponza at 2560 x 1600 on an RTX 5070

Velos - Rendering Hardware Interface

- Designed and implemented a backend-independent RHI with Vulkan and Direct3D 12 backends
- Abstracted pipelines, resource binding, command recording and synchronization
- Automated shader reflection and pipeline-layout generation for SPIRV-V and DXIL

Professional Development

Game Industry Conference (GIC) Sprint
2026 – Katowice, Poland

Attended talks on game engine technology, rendering, and graphics programming.

Languages

German Native

Polish Native

English C1